**Higher National Diploma in Engineering (Electrical)**
**Fourth Year, First Semester Examination – 2021**
**EE 3212 Broadcast Technologies**

Instructions for Candidates:
Answer any **FIVE (05)** questions
All questions carry equal marks

No. of Questions: **06**
No. of Pages : **03**
Time : **(03) Three Hours**

**01.**

i. Explain the term of *Television Broadcasting* clearly. **(03 Marks)**

ii. What are the critical limitations which lead to the development of Digital Broadcasting Technologies? **(03 Marks)**

iii. Differentiate the Analog and *Digital TV.* **(04 Marks)**

iv. Explain *ZENITH-GE* System for FM Broadcasting and draw the frequency spectrum of the composite FM Stereo Signal. **(04 Marks)**

v. Briefly describe the following *recoding methods* using suitable diagrams. **(2x3=06 Marks)**
   a. X-Y Technique
   b. A-B Technique
   c. M/S Technique

**[Total 20 Marks]**

**02.**

i. Describe the term *Matrixing* in FM Stereo broadcasting. **(02 Marks)**

ii. Draw the block diagram of *VHF FM Stereo Encoder* and briefly explain how it works by pointing out the functions of following blocks.
   a. Pre-Emphasis
   b. Matrix
   c. Frequency Doubler **(02 x 3 = 06 Marks)**

iii. Define the *pre-emphasis* and *de-emphasis* in VHF Broadcast **(04 Marks)**

iv. Draw the block diagram of a *Stereo FM Receiver* indicating three major sections (i.e Mono Mode Section, Stereophonic Mode Section and Common Section for both Mono & Stereo Modes) separately. **(05 Marks)**

v. Briefly explain *Cross Talk between L and R Channels.* **(03Marks)**

**[Total 20 Marks]**

**03.**

i. Distinguish *Hue* and *Saturation.* **(03 Marks)**

ii. What is the purpose of *an AGC Circuit* in a TV Receiver? **(03 Marks)**

iii. Define the *Contrast Ratio.* **(03 Marks)**

iv. Why is *De-emphasis* done in TV Receivers? **(02 Marks)**

v. Among the color different signals (G-Y) is not chosen for transmission. Give reasons for this. **(03 Marks)**

vi. Draw a simplified block diagram of *Monochrome TV Camera Tube* and briefly explain its operations. **(06Marks)**

**[Total 20 Marks]**

**04.**

i. Draw the block diagram of the *Color TV Receiver* and explain working principles in details. **(07 Marks)**

ii. List any three key requirements to be satisfied for the compatibility in Television Systems. **(03 Marks)**

iii. What is the *Composite Video Signal*? **(03 Marks)**

iv. What are the functions of *TV Tuner*? **(04 Marks)**

v. In Monochrome Television, what are the three main components in *Composite Video Signal* (CVS) and state the purpose of these three signal components. **(03 Marks)**

**[Total 20 Marks]**

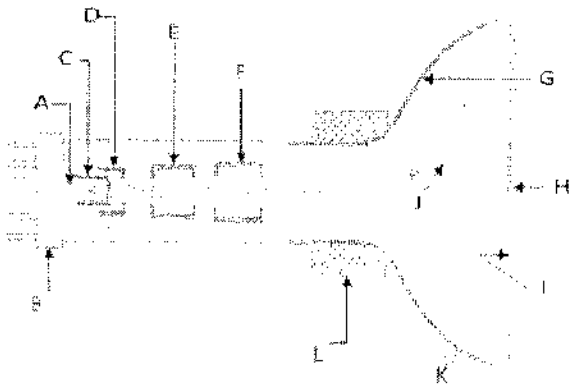
**05.**

i. Why flicker is NOT removed by Progressive Scanning? **(03 Marks)**

ii. Why is *Vertical Scanning* very slow in comparison to *Horizontal Scanning*? **(02 Marks)**

iii. Why is the number of scanning lines in a frame odd? **(04 Marks)**

iv. The following figure shows the *Elements of Picture Tube (B/W)*. State the components name marked by A, B, C, D, E, F, G, H, I, J, K and L. **(06 Marks)**




v. What is the importance of a *Back Porch*? **(02 Marks)**

vi. What is the main function of the *Blanking Pulses*? **(03 Marks)**

**[Total 20 Marks]**

**06.**

i. Give two applications of *Video Tape Recorders*. **(03 Marks)**

ii. What are the drawbacks of *Delta Gun Picture Tube*? **(04 Marks)**

iii. Compare Finite-duration Impulse Response (FIR) Filters with Infinite-duration Impulse Response (IIR) Filters **(04 Marks)**

iv. Differentiate the DVD from CD **(03 Marks)**

v. Write short notes on the following terms. **(02 x 3 = 06 Marks)**

**a)** CATV

**b)** Stereo Tape

**c)** Pulse Code Modulation

**[Total 20 Marks]**